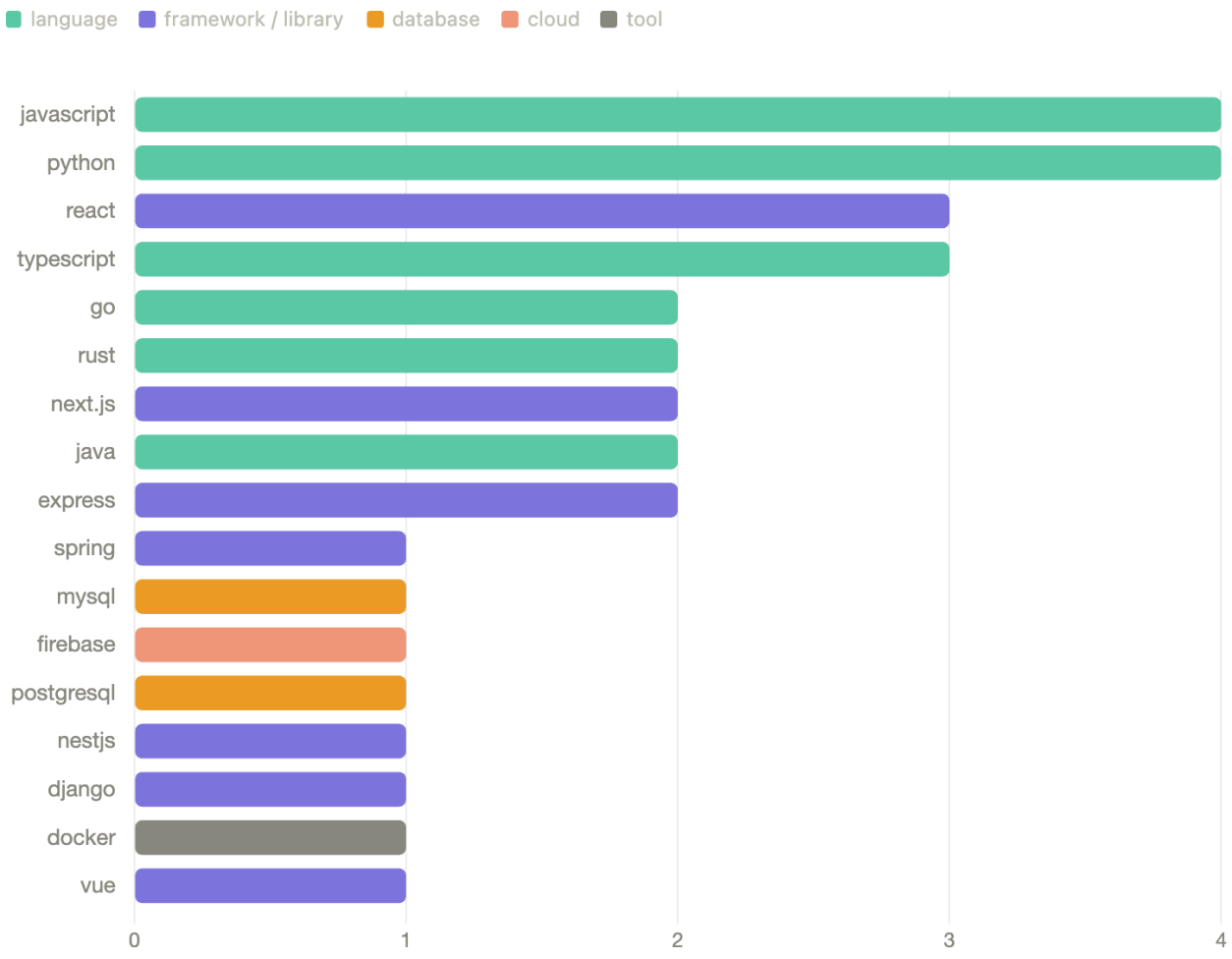
5. Reporting

Reports are generated using direct SQL queries against the PostgreSQL database, combining data across multiple tables using joins and aggregation.

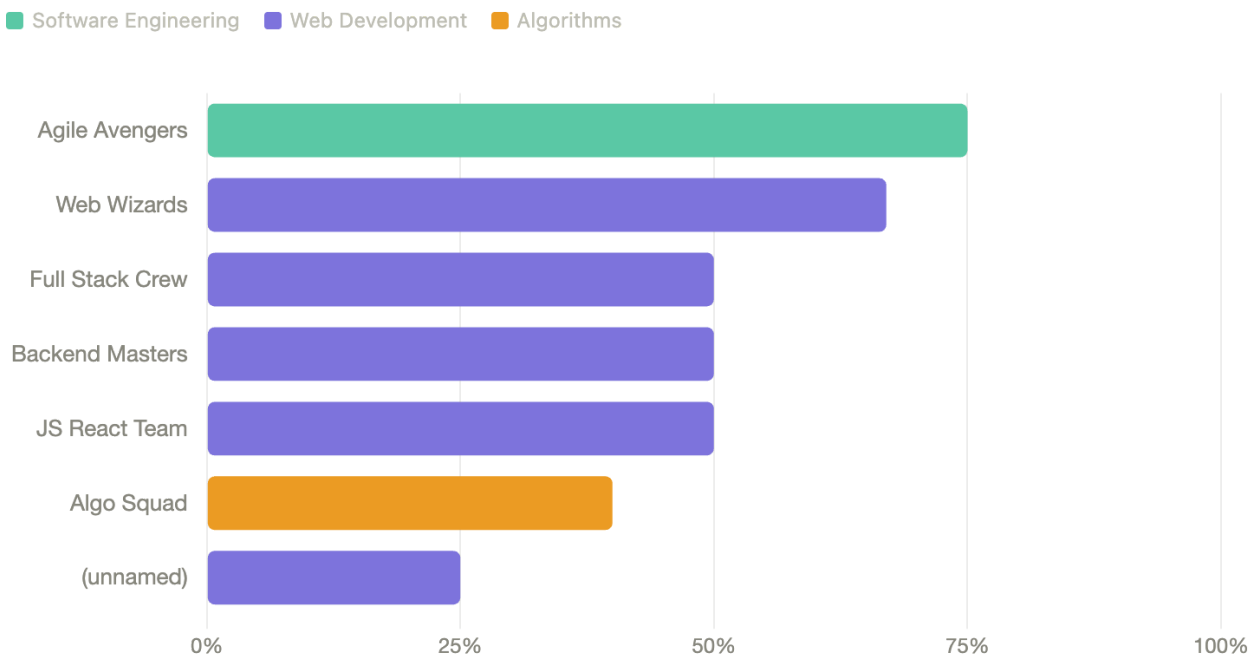
Report 1: Most Common Skills Across Users

```
SELECT s.name, s.type, COUNT(ak.account_id) AS user_count
FROM skills s
JOIN account_skills ak ON s.id = ak.skill_id
GROUP BY s.id, s.name, s.type
ORDER BY user_count DESC;
```

Joins skills and account_skills to count how many users have each skill, grouped by skill name and type. This is useful for understanding what technologies are most represented on the platform. Administrators could use this to see whether the user base skews toward certain languages or frameworks.



Report 2: Group Fill Rate per Class

```
SELECT c.name AS class, g.group_name, g.max_members,
      COUNT(ag.account_id) AS current_members,
      ROUND(COUNT(ag.account_id)::numeric / g.max_members * 100) AS fill_pct
FROM groups g
JOIN classes c ON g.class_id = c.id
LEFT JOIN account_groups ag ON g.id = ag.group_id AND ag.is_pending = false
GROUP BY c.name, g.group_name, g.max_members
ORDER BY fill_pct DESC;
```

Joins groups, classes, and account_groups to calculate how full each group is relative to its max capacity. Only confirmed (non-pending) members are counted. This is useful for students trying to find groups with open spots, and for identifying which classes have the most active group formation activity.